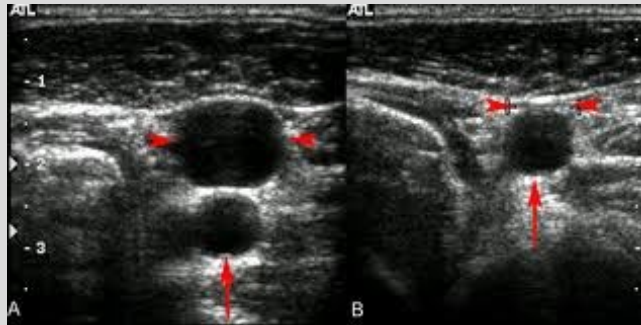
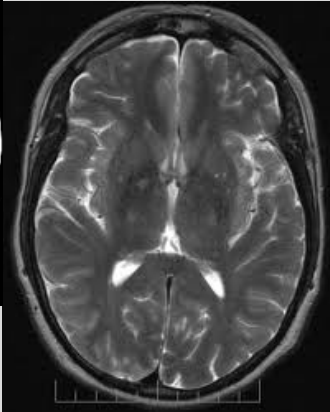


Modalities in Neuroimaging

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- Cross-sectional Imaging
 - CAT scan (CT)
 - Magnetic Resonance Imaging (MRI)
 - Carotid Ultrasound
- Conventional angiogram



CT Scan



- Helical xray beams are directed through the brain, head, neck
- 2D cross sectional images are compiled, and images may be integrated for a 3D view
- Images are displayed in sagittal, coronal and axial planes

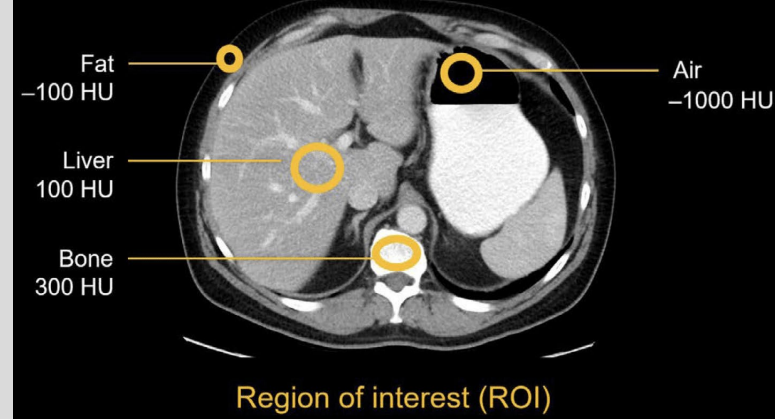
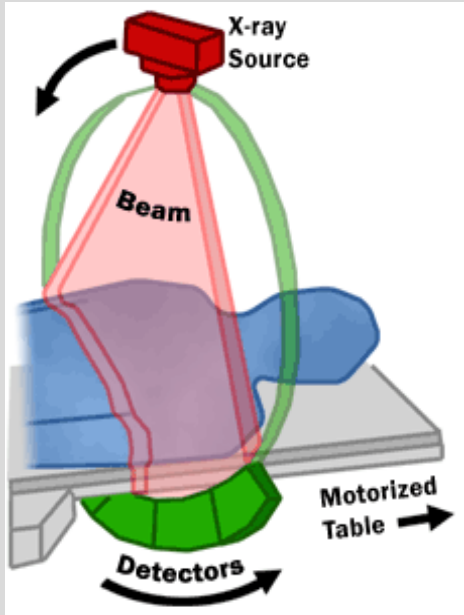


CT

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- Each tissue type **attenuates** the xray beams differently, resulting in different **Hounsfield units**



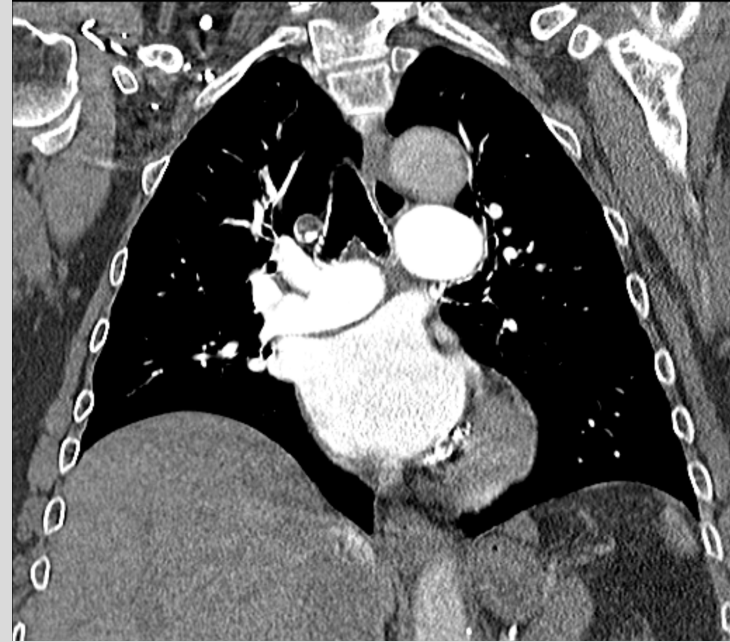
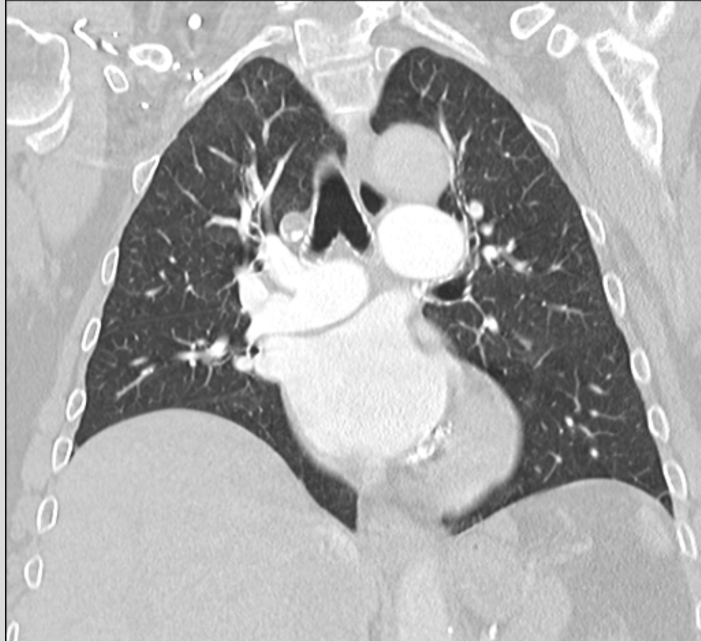
CT: Bone and Brain Windows

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CT: Lung and Soft Tissue Windows



Advantages of CT

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- Superior visualization of:
 - Acute hemorrhage
 - Calcium, Bone
 - Tumor calcifications, fracture
 - Good anatomic detail of multiple tissue types all in one study
- Fast
- Non-invasive
- Relatively cost effective compared to MR



Risks of IV CT Contrast

- Allergic reaction
 - Patients with history of allergy (seasonal, food, previous contrast administration) are prone to having a histamine reaction to contrast
 - Mild, not requiring treatment (5-8%)
 - Feeling of warmth, nausea, itchy throat
 - Moderate (1%)
 - Severe vomiting, hives, swelling
 - Severe (.1%)
 - Life threatening anaphylaxis
 - Management
 - Prescribe contrast with discretion
 - Pre-treatment with Benadryl or Prednisone
 - “crash cart” on site with epinephrine

Risks (Continued)

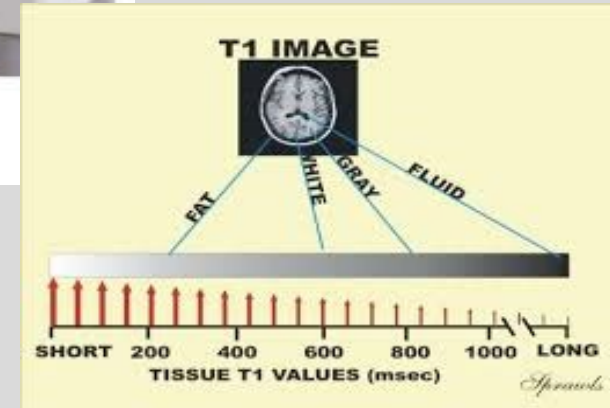
- Nephropathy
 - Patients at risk:
 - Renal impairment (creatinine >1.5)
 - Diabetics
 - Diabetic on metformin - drug must be discontinued for 48hrs
 - Acute renal failure
 - Pre screen for Cr > 1.5 and other pertinent history
 - Schedule hemodialysis within 24 hrs of contrast administration if patient is on hemodialysis
 - Pre-treat with sodium bicarbonate and hydration
- Excretion into breast milk
- Mortality associated with CT contrast 1 in 1.15 million

Magnetic Resonance Imaging

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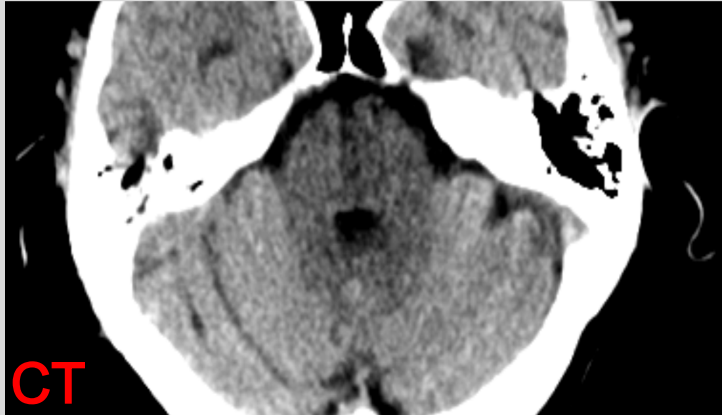
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- Magnetic excitation causes changes in alignment of atoms within the body
- Tissues have distinct patterns of proton excitement and relaxation
- Software measures magnetization vectors that result from the relaxation of protons, producing a 3D image



Advantages of MRI

- Superb visualization of soft tissues including brain, spinal cord, nerves, ligaments, tendons and tumors
- No radiation
- Imaging of vasculature in renal compromised patients, without the need for intravenous contrast



CT



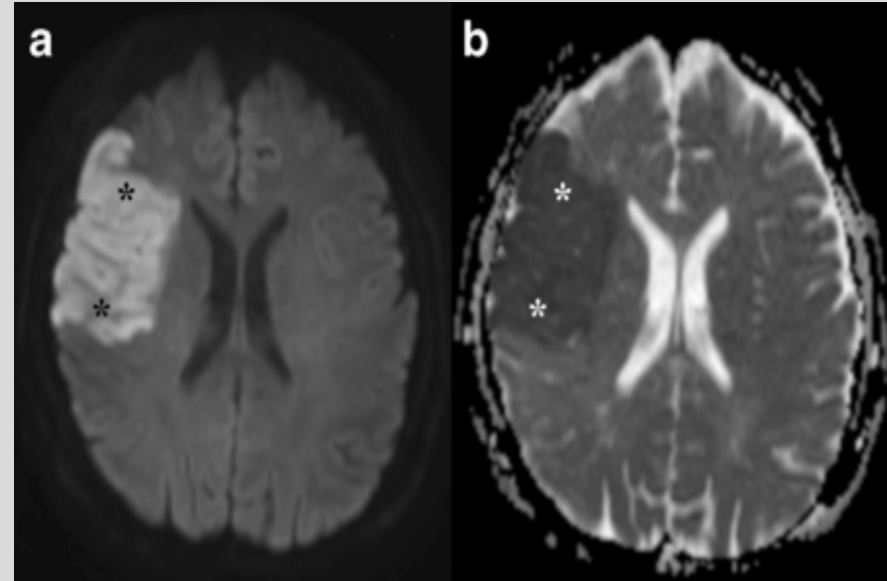
MR

Indications for Brain MRI

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- Early stroke (<12hrs)
 - MRI may be performed after initial CT to detect a CT-occult stroke or to further characterize the extent of ischemia
- Characterization of tumors
- Visualization of cranial nerves
- MR angiography to evaluate stenosis, aneurysm, dissection



Diffusion MRI for stroke

Risks of MRI IV Contrast

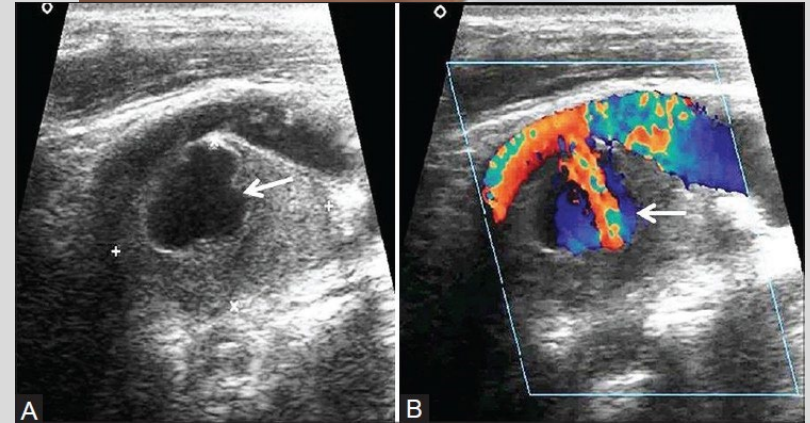
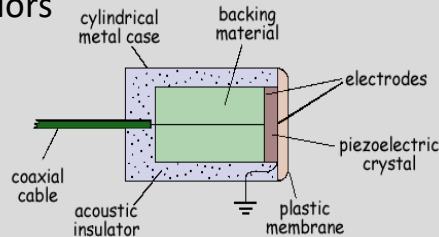
- Nephrogenic Systemic Fibrosis
 - Rare but serious condition characterized by fibrosis of skin and other tissues which can occur hours to months after IV gadolinium contrast administration
 - At risk:
 - Renal impairment, GFR <30
 - Incidence in this population is about 5%
 - Most institutions do not administer to GFR <30
- Gadolinium deposition disease
- Allergic reaction
 - Hypersensitivity - .35%
 - Mild - 1%
 - Anaphylaxis - .01%

Carotid Duplex Ultrasound

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- Step 1 – Grayscale and Color Doppler images
 - Ultrasound transducer is used on the tissue of interest to send and receive sound waves to and from tissue
 - Sound moves differently through various tissues, producing a grayscale image
 - Color doppler - software uses received signal to depict speed and direction of blood flow with colors

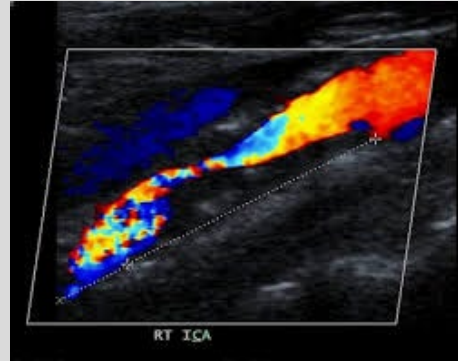


Advantages of Carotid Duplex Ultrasound

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- Non invasive
- Cost effective
- May be performed portably
- Reliable



Conventional Angiogram

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- Fluoroscopy-guided imaging of a vessel of interest by accessing a peripheral vein or artery via catheterization and using intravenous contrast
- Performed by neurosurgeons, interventional radiologists, interventional neuroradiologists, interventional cardiologists

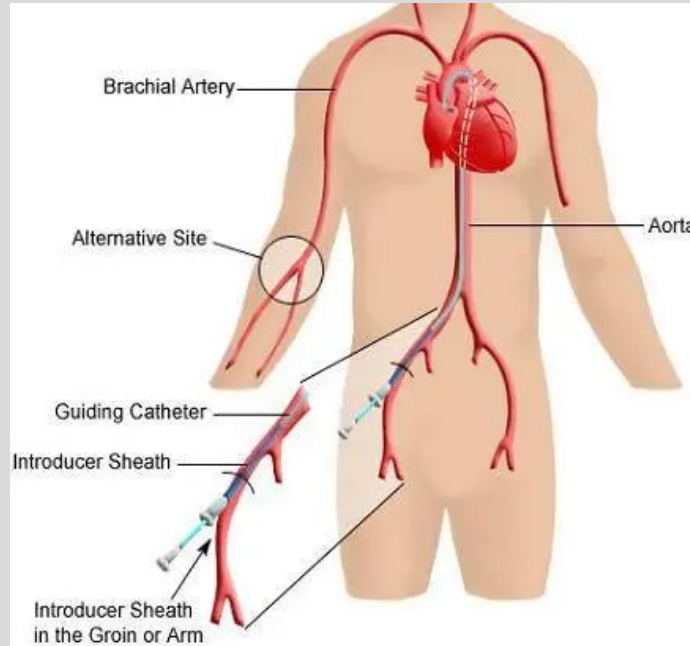


Conventional Angiogram

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- To visualize or map arteries and veins
 - Identify Pathology
 - Treat pathology

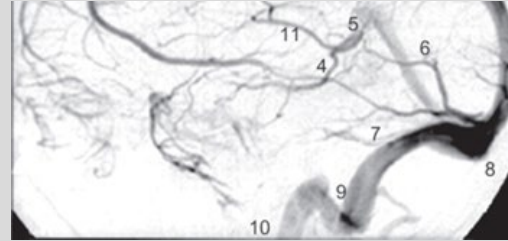


Indications for Conventional Angiogram

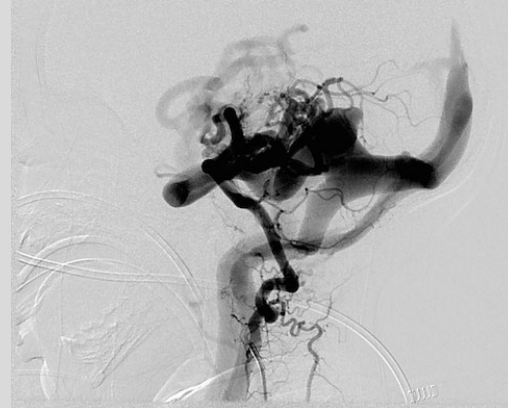
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- Stenosis with intention to treat
 - Congenital
 - Atherosclerotic
 - Compression by external organs
 - Angioplasty
 - Balloon inflation within an area of stenosis
 - Often supported by stent placement
 - New * drug-eluting stents
- Traumatic vascular injury
- Aneurysm, arteriovenous malformation
 - Coil embolization
 - Plugs
 - Gel embolization



Normal cerebral venogram



Abnormal cerebral
venogram with
Diagnosis - arteriovenous
malformation

Summary

- CT BRAIN
 - Best for acute trauma, intracranial hemorrhage
- MR BRAIN
 - Best for evaluation of nerves, ligaments, and when trying to avoid radiation to the patient
- CT BRAIN/MR BRAIN with INTRAVENOUS CONTRAST
 - To evaluate vascular pathology and patterns of enhancement in other lesions
- CAROTID DUPLEX STUDY
 - Cervical
- CONVENTIONAL ANGIOGRAM
 - Best for vascular pathology and cases in which diagnosis and treatment may be paired
 - Known aneurysm, stenosis